Gazi Fahim Abrar

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EDUCATION

University of Minnesota (Twin Cities) Minneapolis, MN | Graduated May 2026
B.S. Computer Science | Concentration: AI, Software Engineering & Robotics | GPA: 3.6/4.0 | Dean's List Fall '23 & Spring '24

Coursework: Advanced Data Structures & Algorithms, Artificial Intelligence, Computational Robotics, Cyber Security, Computer Networking, Data Mining, Operating Systems, Probability & Statistics, Computational Linear Algebra

EXPERIENCE

UMN Networking Lab - Research Assistant Sept 2025 – Present
Networking & AI Research | Advisor: Prof. Zhi-Li Zhang *Minneapolis, MN*

- Architected **RAG** (Retrieval-Augmented Generation) systems for **Text-to-SQL** generation via **Model Context Protocol (MCP)**; engineered prompt engineering pipelines and schema mapping to enable LLMs to generate and execute validated SQL queries on dynamic datasets.
- **1st Place, AT&T AI Telco Troubleshooting Challenge (Track 3):** Outperformed 1,200+ global teams; fine-tuned **Small Language Models (SLMs)** for cloud-edge deployment, reducing inference latency for real-time network fault diagnostics. Presented at **MWC 2026, Barcelona**.

UMN Teen & Families Lab - Research Assistant Sept 2025 – April 2026
Computational Social Science | Advisor: Prof. Xiaoran Sun | Collaboration: Stanford University *Minneapolis, MN*

- Led **BERT**-based sentiment analysis on 100k+ mobile screenshots; built a custom **NLP** pipeline to classify emotional health trends across a longitudinal dataset; co-authored a paper currently under peer review at a **Nature** journal.
- Fine-tuned transformer models (**PyTorch**) on the **GoEmotions** dataset for 28-class multi-label emotion classification on social media data; engineered features with **Pandas** and optimized via transfer learning.

UMN School of Physics & Astrophysics - Learning Assistant Jan 2024 – April 2024
PHYS 1302 (Electromagnetism) *Minneapolis, MN*

- Facilitated technical understanding of Electromagnetism and Maxwell's Equations for large student cohorts; bridged theoretical concepts with practical applications through demonstrations and one-on-one mentorship.

PROJECTS

Senku: Self-Supervised World Model for Planning from Pixels 2026 | github.com/gzee3184/senku_wm | Open-source (MIT)

- Designed and open-sourced **Senku**, a 15.5M-parameter **JEPA**-style **self-supervised world model** that learns from pixels and actions only (no rewards, reconstruction, or pretrained encoder); encodes each frame as a global latent plus 49 spatial state tokens, with a READ/WRITE predictor, auxiliary-predictor gradient split, and **SIGReg** regularization; trained in **PyTorch** on a single GPU.
- Beat the LeWM baseline in closed-loop **model-based planning** on **OGBench Cube**: 88.8% vs. 74.2% success under LeWM's protocol (+14.6 pts, paired p=2×10^{-12}; 82.4% vs. 67.0% on identical expert data), 37.3% vs. 9.0% on grasp-requiring starts, and 68.7% vs. 20.7% on H-JEPA's 50 Cube tasks; matched LeWM on PushT (92.0% vs. 90.2%, not significant).
- Built the **evaluation framework**: held-out start splits, planner chosen on separate validation starts (CEM variant and gradient planner), paired bootstrap/McNemar tests, leave-one-out ablations, and latent probes; released code, per-episode results, and checkpoints on GitHub and Hugging Face.

LongHorizonWM: Subgoal Chaining for Long-Horizon Latent Planning 2026 | github.com/gzee3184/LongHorizonWM

- Designed a subgoal-chaining framework that fixes long-horizon failure in closed-loop latent planning; extended a **JEPA**-style frozen world-model video encoder with a novel curvature regularizer, then trained a chainer and reachability head atop it to convert distant goals into reachable subgoals for a **CEM**-based planner.
- Improved closed-loop planning success on the **PushT** benchmark from a 12.25% baseline to 74.125% at a 100-step goal offset, a statistically validated gain over the official chainer baseline, with consistent improvements across two additional benchmarks.

Graduate Career & Salary Prediction Oct – Dec 2025

- Analyzed IPUMS Higher Ed dataset (461k+ rows, 357 features); applied **EDA**, categorical encoding, and Logical Skip imputation to handle sparse survey data at scale.
- Achieved 72.9% accuracy via **XGBoost** / **Random Forest** / **CatBoost** ensemble; used **Mutual Information (MI)** for feature selection and **SHAP** for model explainability on career outcome prediction.

ROS2 Hand Gesture Robotics Control Nov 2025

- Built a **ROS2** package integrating **MediaPipe** and **OpenCV** for real-time hand landmark tracking; translated gestures into Twist velocity commands for differential drive robots, validated in **Gazebo** simulation.

SKILLS

Languages:	Python, C++, C, Java, SQL, JavaScript, HTML/CSS
Technologies:	PyTorch, TensorFlow, MuJoCo, OGBench, Hugging Face, Docker, Git, ROS2, OpenCV, MediaPipe, NVIDIA NIM, OLLAMA, XGBoost, CatBoost, SHAP, Pandas, Jupyter, Gazebo, Arduino
Concepts:	World Models, JEPA, Self-Supervised Learning, Model-Based Planning (CEM, Gradient), Latent Planning, Reinforcement Learning, Video/Computer Vision, Evaluation Frameworks, Fine-Tuning, Transfer Learning, BERT, NLP, RAG, SLMs, MCP, Mutual Information, EDA, Data Structures & Algorithms